Installing R and Python

Intro

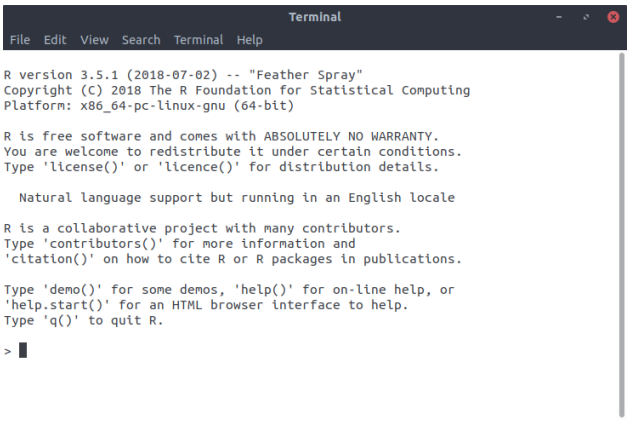
This document will walk you through various pieces of software that you need to download in order to get up and running with R and Python for the course. For R, this includes installing the R language itself, as well as a friendly environment (RStudio) in which to write your R code (this is called an "integrated development environment" or IDE for short). In addition, we ask that you also have a LaTeX installation on your computer (for writing fancy math) and that your R environment is capable of producing RMarkdown documents (these are a great way to do your homework). For Python, this document will cover the installation of Python using Anaconda. Anaconda is a platform that simplifies package management and you are encouraged to install Python through the use of Anaconda to simplify the installation process as Anaconda comes pre-installed with a lot of data science libraries that we will be using throughout the course. We will also guide you to install Spyder, which is the recommended IDE for Python (you could also use Jupyter Notebook if you prefer, but installation is not covered in this document). It is, however, discouraged to install R and RStudio through Anaconda as the R installation through Anaconda will put R under the Anaconda environment and would require additional path setting changes so that LaTeX would work under this environment.

R

R is software for data analysis. It's like Python, but geared toward statistical work; it's like Stata, but better for programming; it's like Matlab, but free.

You need to download and install R onto your computer. R is hosted on the "Comprehensive R Archive Network" (aka CRAN). It's not the prettiest website out there. To download R, go to `https://cran.r-project.org/` and click on the download link for your operating system (Linux, Mac OSX, or Windows). Depending on your OS, you'll be taken to a second page where you have to choose which "version" or "package" to download. Linux users should choose their flavor of Linux (Debian, Ubuntu, etc.); Mac users should pick the top link (R-4.0.2.pkg at the time this document was created); Windows users should choose "Base". You might have to click through one more page – just get the recommended version for your OS. After R downloads, install it as you would any other piece of software.

Once you've installed R, open up the application.[1] It will be basic. You'll get one floating window called the "R Console" or "R Terminal" with a bunch of text and a command prompt.



[1] Linux users can run R in the terminal by executing the terminal command "R" without quotes.

At the prompt (which looks like a greater-than sign, >) type `2+2` and hit enter. R will talk back to you, telling you `4`, like so:[2]

```
2+2

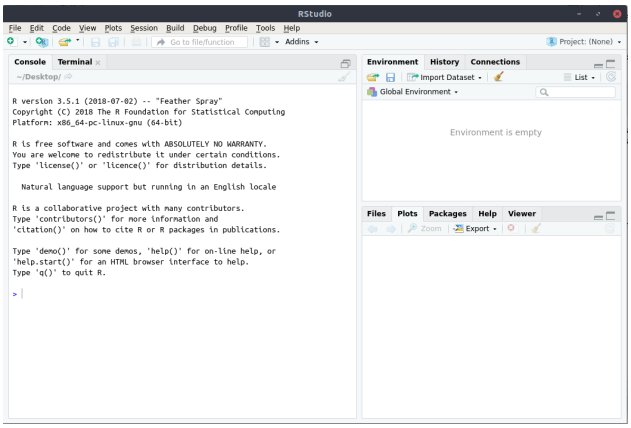
## [1] 4
```

This works, but it's a pretty unfriendly way to use R, so close the R application. To make the experience with R friendlier, you'll want an IDE. My favorite IDE for R is RStudio. You're welcome to use any IDE that you like, but this document will be specific to RStudio. (You'll have to figure out the rest of this on your own if you go with another IDE Visual Studio or an extensible text editor like Atom.)

RStudio

You need to download and install RStudio. To do so, go to `https://www.rstudio.com/`. At the top, click on "Products" and then "RStudio". Choose "Desktop" and then "Download RStudio Desktop". Once downloaded, install it.

Now open up RStudio. It should find the R application you previously installed automatically. Your now running that application (R), but now with a much prettier interface (RStudio):



On the left side of the screen in a window called "Console", again type `2+2` and hit enter. R will return `4`.

Next, type:

```
x = 3+7
```

Now you'll see the object `x` in the "Environment" window on the top right. Go back to the console on the left and type `x` and hit return. R will return `10`.

RMarkdown

Let's say we don't want to just write R code, but we want the code, some results from our code, and some floating text (like paragraphs explaining what the code does) all to show up in the same document (i.e., just like this document!). RMarkdown is a way to make that kind of document.

Unlike what-you-see-is-what-you-get programs like Microsoft Word, with RMarkdown you "code" your document, including the document's contents and its formatting. That means, for example, that you write "

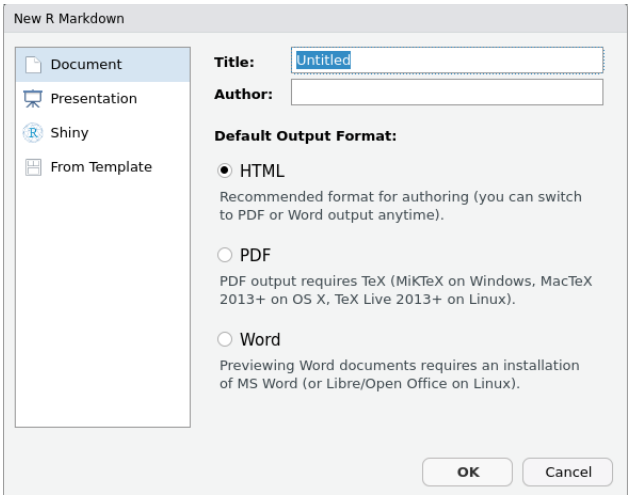
[2] The [1] indicates that the result is a vector (R doesn't have scalars) and that the first element of that vector is 4.

beta" and when you compile the document, you see "β". Or, as another example, you code _word_ in order to set "word" in italics, like so: *word*. For a little intro on the RMarkdown language, check out this site: `https://rmarkdown.rstudio.com/lesson-1.html`.

You don't need to install anything for RMarkdown; it's built into RStudio. In RStudio, click on the dropdown arrow next to the icon for a new document (this is at the top left of RStudio), then choose "RMarkdown". The first time you do this RStudio will tell you that it needs to install packages, say yes/ok and let those install.

If there is any error with this step, go to the bottom right window in RStudio and click on the tab called "Package". Then click "Install". Type "rmarkdown" (all lowercase and without the quotes) and click "Install".

Once you've got the packages installed, go back to the blank page icon at the top left and create a new RMarkdown document. A diaglog box will pop up asking for you to provide a document title and author, it also gives you the option to select "html" or "pdf", among other things.



Right now, you have enough software to make an html document, but you don't yet have enough to create a pdf. For that, you need Latex. Hit "Cancel" and quit RStudio; let's go get LaTeX.

LaTeX

This can get complicated fast, so let me try to provide you with just the right information. TEXis a computer program for typesetting documents. Whereas RMarkdown is basic and easy to use, TEXis super-unfriendly to use. But, TEXis like Microsoft Word for people who write math – everybody in academia uses TEX– so we need to learn how to use it as well.

Thankfully someone made LATEX(pronounced la-tech). LATEXis a set of macros for TEXthat takes the experience of working with TEXfrom horrible to somewhat-manageable (there is no good/fun way to do TEX/LATEX). You need to install LATEXand doing so will automatically install TEXfor you. LATEXhas a bunch of different "flavors" named MacTex or MikTex – they're all basically the same.

I don't remember all the steps to install LATEXand they vary based on your OS. So, for Mac users, google "install MacTex". For Windows users, google "install MikTex". For Linux users, your distro likely already has LATEX, but if not, type the following into the Bash shell: `sudo apt-get install texlive` (or replace

apt-get with your package manager). Regardless of which OS you're on, this should be a simple process – just download and install like anything else.

We'll be using LaTeXas an intermediate step when creating pdf documents. In fact, if we do everything right, you may not even notice that LaTeXis being used. However, if you want to write LaTeXdocuments and compile straight to pdf (without any R or Rmarkdown), you can download an editor for LaTeX(this is like how we use RStudio for R). I use TexMaker on Windows, but there are lots of options, and you should feel free to get whichever you like (or none of them).

Creating Your First RMarkdown

Let's see if we can get everything working, and to test that, we'll use the RMD version of the homework template.

On your computer, create a new directory (also known as a folder) and put the RMD version of the homework template in that directory. Open RStudio (if you closed it earlier). Go to RStudio's "Tools" menu and click "Global Options". Click "Sweave" on the left and then change the top dropdown from "Sweave" to "Knitr". Then exit global options.

Open the homework template RMD file. A new window should open in RStudio on the top-left. Click the "Knit" button in the top-middle (on newer versions of RStudio, this will say "Compile PDF" instead of "knit"). A new tab will open near the "Console" in the bottom-left called "RMarkdown". If everything works, in a couple seconds, a PDF will pop up. If not, there will be an error message in the "RMarkdown" tab. If you get an error, Google that error and try to fix your problem. Again, we recommend that you do not install R and RStudio through Anaconda as you would require to change the R environment setting for the LaTeXto work.

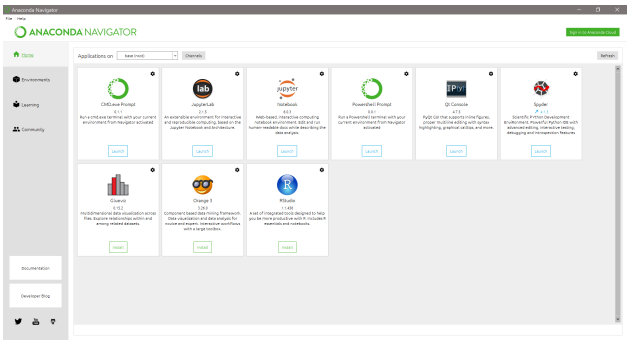
For any other problems, contact your TA.

Installing Python

Python is a general-purpose object-oriented programming language. Due to the exhaustive collection of libraries, Python has been popular among the data science community.

We recommend that you download and install Python through Anaconda. Anaconda provides a platform that simplifies package management and allows you to launch popular data science Python IDEs such as JupyterLab, Jupyter Notebook and Spyder from the platform. To download Anaconda, go to `https://www.anaconda.com/products/individual` and click on the download link for your operating system (Linux, Mac OSX, or Windows). After finishing the download, install Anaconda just like any other software.

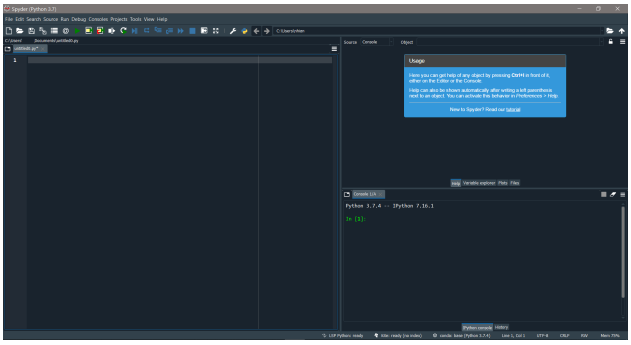
Once you've installed Anaconda, open up the Anaconda navigator and you'll get a GUI with a bunch of applications that are ready to be launched/installed.



Anaconda will automatically install Python in the Anaconda environment, but you'll want an IDE to make it easier to code. My favorite IDE for Python is Spyder and it has a very similar layout to RStudio. And again, you're welcome to use any other IDE that you like but this document is specific to Spyder.

Spyder

You first need to install Spyder and the Anaconda Navigator makes it really easy to install Spyder. We recommend that you install Spyder 4 as the 4th version provides a graphic panel that makes plotting very interactive and easy. You can install a specific version of Spyder by clicking on the gear icon on the top-right of the Spyder "card" in the Anaconda Navigator. Once installed, you can now launch Spyder from Anaconda by clicking the Launch button.



On the bottom right side of the screen there is the console window and you can type in commands to the console window. For example, if you type in `2+2` and hit enter, Python will return `4`. On the top right side of the screen there are multiple panels that are available by default and the Plots panel is available by clicking on "Plots".

Next, type in the console:

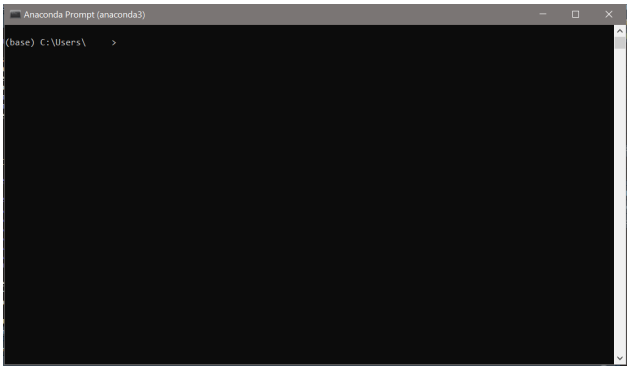
```
x = 3+7
```

Now you'll see the object x in the "Variable Explorer" window on the top right. Go back to the console on the type x and hit return. Python will return 10.

By installing Python through Anaconda, your environment comes pre-installed with most of the fundamental data science libraries such as `NumPy`, `pandas` and `matplotlib`. For example, to import the `NumPy` library you can type in `import numpy` in the console. Then, you can check your `NumPy`'s version by typing in `numpy.__version__` and Python will return the installed `NumPy` version in your Anaconda environment.

Installing Libraries

Installing libraries in the Anaconda environment can be done mainly through two ways: the Anaconda prompt and using the Anaconda Environment Navigator. Open source libraries are available on the Anaconda Cloud which is the Anaconda repository. The Anaconda prompt can be launched by clicking on the CMD.exe Prompt on the Anaconda Navigator (or by typing Anaconda Prompt on the Windows search bar for Windows) and libraries can be installed through the `conda install PackageName` command. For those who are familiar with Pythons `pip install PackageName` command, the `conda install` command does the same with the additional benefit that it takes care of packages dependencies automatically. An example, of the Anaconda prompt is shown below and it looks exactly like Windows command prompt:



We recommend that you update these important libraries to the newest version through either the command prompt or navigator. If using the command prompt, you can update the libraries by running `conda install PackageName` in the prompt. If you are using the navigator you can go to the "Environments" tab on the left bar of the navigator and under the `base (root)` environment, search for the packages that you want to update and update them.

- `numpy`
- `pandas`
- `matplotlib`
- `statsmodels`
- `scipy`
- `sklearn`

These libraries will be used extensively throughout the course and the code snippets provided will be using the latest version of these libraries at the time of writing. Furthermore, we also recommend that you install the `rpy2` library by entering the command `conda install rpy2` in the Anaconda command prompt. This library will allow us to run R commands from Python and thus we would be able to load various R datasets that are used for examples for the course. For more package management and for a quick guide on how Anaconda works, I have provided the following user guide `https://conda.io/projects/conda/en/latest/user-guide/getting-started.html` that will give you a quick introduction on Anaconda.